

PoWoBD: A Bangla Dataset and Ensemble Transformer Framework for Political Barriers Classification Against Women in Bangladesh

Abstract—Women’s participation in political leadership in Bangladesh remains disproportionately low, and social media harassment is one of the barriers discouraging it, yet no dataset exists that categorizes this harassment with the specificity needed to study or moderate it. This paper introduces PoWoBD, a survey-grounded Bangla dataset of 23,532 Facebook comments annotated across seven fine-grained categories derived from an initial nine-category taxonomy of political barriers faced by Bangladeshi women, including personal attacks, character shaming, gender discrimination, and safety threats, derived from a survey of over 100 Bengali women and annotated by a demographically diverse team with substantial inter-annotator agreement (Fleiss’ $\kappa = 0.71$). We evaluate the dataset under three progressively broader label schemes, showing that classification accuracy improves as semantically overlapping fine-grained categories are consolidated: from an unstable nine-class baseline, to 68.49% accuracy on a four-class scheme, to 82.12% accuracy on a three-class positive/negative/neutral scheme using a fine-tuned BanglaBERT model. A soft-voting ensemble of BanglaBERT, BanglishBERT, and XLM-RoBERTa further improves this result to 83.76% accuracy and a macro F1 of 0.8160, the best result across every configuration tested. Beyond its modeling results, PoWoBD offers a fine-grained resource for content moderation, discourse analysis, and policy-facing research on gender-based political harassment in low-resource language settings.

Index Terms—component, formatting, style, styling, insert

I. INTRODUCTION

Women comprise half of the population of Bangladesh. They are making progress in several sectors yet facing discrimination in many aspects of their lives. It is important to have women leadership in political fields to speak against these inequalities and reduce the unfairness towards women rights.

At the same time women are underrepresented for political leadership roles. Their participation in the political zone is significantly low because of many barriers. Social media has a big influence over this issue. Online harassment is one of the most disturbing problems that women have to encounter. They face a large amount of hate speech, gender-based trolling, character assassination and online abuse in the form of online comments in numerous social media platforms. This kind of digital harassment

discourages women from participating in any political leadership roles.

As a result, it is vital to make a secure cyber space. Deep learning models can be used for this task. These models can be applied to auto detect abusive comments, hate speech and filter them in order to construct a personalized cyber space. These deep learning models need properly labeled data. However, There is no properly defined and categorized dataset which can analyze those hate comments based on their respective criteria. In recent works, BERT [1] become popular in sentiment classification task. Most of the research papers fine-tuned different variant of banglaBERT [?] and multilingual-BERT [?] for sentiment analysis in Bangali language for their high performance [2]–[6] but the dataset they have used only contains tri-class classification [3], [7]. Moreover, other papers have also used a dataset which contains a binary classified dataset. These papers mainly focus on generally detecting the positive, negative and neutral comments but they are not appropriately classifying them. The lack of a multi categorized dataset has turned out to be a significant issue which is creating a research gap between the analysis and the technical solution.

To mitigate this issue, we constructed a survey-based, multi-classified dataset. We first conducted a survey to identify the categories of the dataset. After identifying the labels from the survey, we manually annotated comments according to the respective labels. This approach ensures that the research can identify and classify online speech at a level of granularity existing Bangla resources do not offer. After constructing the PoWoBD dataset, we evaluated it under three label schemes and show that classification accuracy improves as semantically overlapping fine-grained labels are consolidated, from an unstable nine-label baseline to 82.12% accuracy on the three-class (positive, negative, neutral) scheme with a fine-tuned BanglaBERT model. Finally, out of four transformers – mBERT, BanglaBERT, BanglishBERT, and XLM-RoBERTa – we combined the top three into a soft-voting ensemble, reaching 83.76% accuracy and a macro F1 of 0.8160. In summary, this paper’s contributions are:

- PoWoBD, a survey-grounded, Bangla dataset targeting political barriers against women;

- an analysis of how label granularity affects classification reliability for this task; and
- an ensemble transformer framework that achieves the best reported result across the label schemes tested.

II. RELATED WORK

A. Bangla Sentiment Analysis

BLP-2023 Task 2 built a three-class Bangla sentiment benchmark from MUBASE and SentNoB, evaluated by Micro-F1 across mBERT, BanglaBERT, XLM-RoBERTa, and few-shot mT5/GPT-3.5 [3]. EmptyMind fine-tuned transformers within this task on Facebook and Twitter data [4]. SentNoB covered noisy social-media text but reported weak annotation agreement (Fleiss’ $k = 0.53$) [5]. SentiGOLD used a larger, five-class, 30-domain corpus with much stronger agreement ($k = 0.88$) [8]. BanglaBook’s 158,065 book reviews carry severe class imbalance [7]. Outside Bangla, an Indonesian stock-sentiment corpus [4] and a 22-domain Weibo dataset [6] do not transfer across languages. Hawkes-Graph modeled temporal sentiment propagation with graph neural networks on its VISTA dataset, using propagation-specific metrics that are hard to compare against standard classification benchmarks [2]. Domain-restricted studies include Naive Bayes sentiment on TripAdvisor and Rotten Tomatoes reviews [13] and CNN/RNN-based cricket-domain sentiment [15]; neither generalizes past its own domain.

B. Hate Speech Detection

BengaliSent140 combined seven datasets into 139,792 binary-labeled samples. BERT performed best among the models tested, though the binary scheme loses the finer target and type information the source datasets had [10]. Retrivr, built for BLP-2025 Task 1, used transformer ensembles with multi-task learning and reached roughly 72–73% Micro-F1 [11]. HateBertBN paired BanglaBERT embeddings with CNN, LSTM, and MLP heads, trained on a fairly small manually annotated set [12]. BanTH added multi-label, seven-category annotation, but only for transliterated Romanized Bangla [16]. Das et al. built a code-mixed corpus of 10,178 tweets, which is small by dataset standards [17]. BD-SHS used a hierarchical three-level labeling scheme but tested it only with SVM and Bi-LSTM [18]. Davidson et al.’s early work established the hate-versus-offensive distinction using a Hatebase-seeded lexicon and Logistic Regression/Naive Bayes, but it is English-only and lexicon-based, so it does not transfer directly to Bangla [14].

C. Women-Focused NLP

WoNBias is the closest prior work to this study: a 31,484-instance Bengali dataset for gender-based bias and prejudice, fine-tuned with BanglaBERT and evaluated through recall, accuracy, and error analysis. Its main limitation is isolation. Gender bias appears elsewhere only as one minor category inside broader hate-speech taxonomies, not as its own dedicated task [7].

D. Political Perspectives

Salam et al. used BDHS 2022 data with machine-learning models to find predictors of women’s empowerment, but the analysis is correlational, not text-based [19]. Ahmed [20] and Ara and Northcote [21] documented structural and institutional barriers qualitatively. Both establish the social relevance of the problem without applying any computational or NLP methods [20], [21].

Collectively, this literature reveals three trends motivating the present study. First, Bangla sentiment and hate-speech research has expanded through two complementary directions: dataset aggregation to increase training scale and harmonize heterogeneous labels [10], and multi-label or hierarchical taxonomies to represent target groups and types of harmful content more precisely [16], [18]. Second, gender-targeted textual bias detection remains comparatively underexplored relative to general sentiment and hate-speech classification – WoNBias directly addresses prejudice against women [7], but broader datasets such as BanTH, BD-SHS, and code-mixed Bengali corpora include gender-related categories as one component within wider harmful-content taxonomies [16]–[18]. Third, evidence from cross-lingual and cross-domain studies confirms that annotation inconsistency, linguistic noise, domain shift, and class imbalance remain persistent challenges in sentiment and harmful-language analysis [4], [6], [13], [15]. These gaps – particularly the limited availability of fine-grained, rigorously error-analyzed Bangla resources for gender-related bias in a political context – directly motivate the approach adopted in this work.

III. DATASET CONSTRUCTION

In this section, we comprehensively describe the methodology of our Dataset, Which is broadly categorized into i) Task Definition, which is an overview of the background Study of What types of political barriers are faced by Bengali Women Politicians, ii) Taxonomy Design, Which represents the initialization of our research work flow to prepare our dataset, iii) Data Collection, contains information about Data collection and processing for final Dataset creation, iv) Annotation Protocol, this section represents data labeling, annotator introduction & evaluations v) Finally, the Dataset Statistics part ensure the dataset quality. So, let’s discuss every section in detail:

A. Task Definition

One of the most important issues for socio-economic development in the current world is gender equality. Compared to developed countries where gender inequality has been reduced across all sectors, middle-income or developing countries like Bangladesh face much higher gender inequality. For this reason, women’s participation must be ensured in political leadership, starting from education. Although women’s participation in other sectors such as

Table I
METHODOLOGICAL COMPARISON OF KEY EXISTING STUDIES AND THE PROPOSED SYSTEM.

Study	Domain / Task	Best Model / Reported Result	Gender & Political Focus
BLP-2023 Task 2 [3]	General Bangla sentiment	mBERT, BanglaBERT, XLM-R, few-shot mT5/GPT-3.5 (Micro-F1)	No
SentNoB [9]	Noisy social-media sentiment	Baselines through fine-tuned mBERT; $\kappa = 0.53$	No
SentiGOLD [8]	Multi-domain sentiment	N/R accuracy; $\kappa = 0.88$	No
BanglaBook [3]	Book-review sentiment	N/R; evaluated via F1, weighted F1	No
BengaliSent140 [10]	Hate-speech detection	BERT (highest accuracy among models tested)	No
BanTH [16]	Transliterated hate speech	Multiple pretrained LMs benchmarked; N/R	Partial
WoNBias [6]	Gender-based bias, harmful language	Fine-tuned BanglaBERT; recall/accuracy via confusion matrix	Gender only
PoWoBD (Ours)	Political, gender-targeted harassment	Soft-voting ensemble: 83.76% acc., macro F1 0.8160	Gender & political

N/R = not reported in the cited study.

education, healthcare, and the workplace can be increased relatively quickly, ensuring that their participation in political leadership is not as rapid. Even if women are willing to assume political leadership, they can still face obstacles in participating. For this purpose, our research attempts to identify the types of barriers Bangladeshi women encounter in entering political leadership and how to reduce those barriers, and how to ensure the safety of bangali women politicians from Social Media harassment and cyber insecurity by using the AI/ML model. But The lack of structured dataset on this particular problem is a key challenge to train the model properly. So We are addressing this issue to solve by creating a PoWoBD dataset on social media comments.

B. Taxonomy Design

In this study, to uniquely classify PoWoBD dataset comments, we needed to design a taxonomy that helps classify each comment with a unique classification label. To identify these classification labels, we conducted a survey among Bangladeshi women. To this end, we initially collected women’s direct or indirect experiences regarding the kinds of obstacles women face in entering political leadership. For this purpose, we gathered responses by asking two questions to Bengali women. The questions are:

- 1) “আপনি যদি পরোক্ষভাবে (বন্ধু, পরিবার, সহপাঠী, সংবাদ, সোশ্যাল মিডিয়া বা সমাজে দেখা) অভিজ্ঞতা হয়ে থাকে, তাহলে এক বাক্যে শেয়ার করুন মেয়েরা রাজনৈতিক নেতৃত্বে আসতে কী ধরনের বাধা ফেস করছে।” (If you have experienced it indirectly—from friends, family, classmates, news, social media, or society—share in one sentence what kind of obstacles women face in entering political leadership.)

- 2) “আপনি যদি পরোক্ষভাবে (বন্ধু, পরিবার, সহপাঠী, সংবাদ, সোশ্যাল মিডিয়া বা সমাজে দেখা) অভিজ্ঞতা হয়ে থাকে, তাহলে এক বাক্যে শেয়ার করুন মেয়েরা রাজনৈতিক নেতৃত্বে আসতে কী ধরনের বাধা ফেস করছে।” (If you have experienced it directly (in your own life), share in one sentence what kind of obstacles women face in entering political leadership.)

After collecting 100+ responses from Bengali women of various ages and professions, we analyzed the survey responses and derived the class labels. After extracting the labels, we identified a total of nine class labels from the survey. The extracted class labels with descriptions are presented below:

- 1) Personal Attack: Insulting, abusing or humiliating an individual directly.
- 2) Character Shaming: Attacking a person’s character, morality or religious identity.
- 3) Family Restriction: Family barriers or pressure, not getting permission or support from the family.
- 4) Gender Discrimination: Gender-based discrimination or patriarchal mindset.
- 5) Personal Safety Threat: Insecurity, threats, harassment or eve-teasing.
- 6) Psychological Impact: Mental stress, depression or loss of self-confidence.
- 7) Economic Barrier: Financial problems or lack of necessary funds.
- 8) Supportive: Support or positive attitude towards women’s political leadership.
- 9) Neutral: Neutral attitude or no specific opinion about women’s political leadership.

Finally, after obtaining all nine labels, we moved forward to data collection. Two categories, *Family Restriction* and

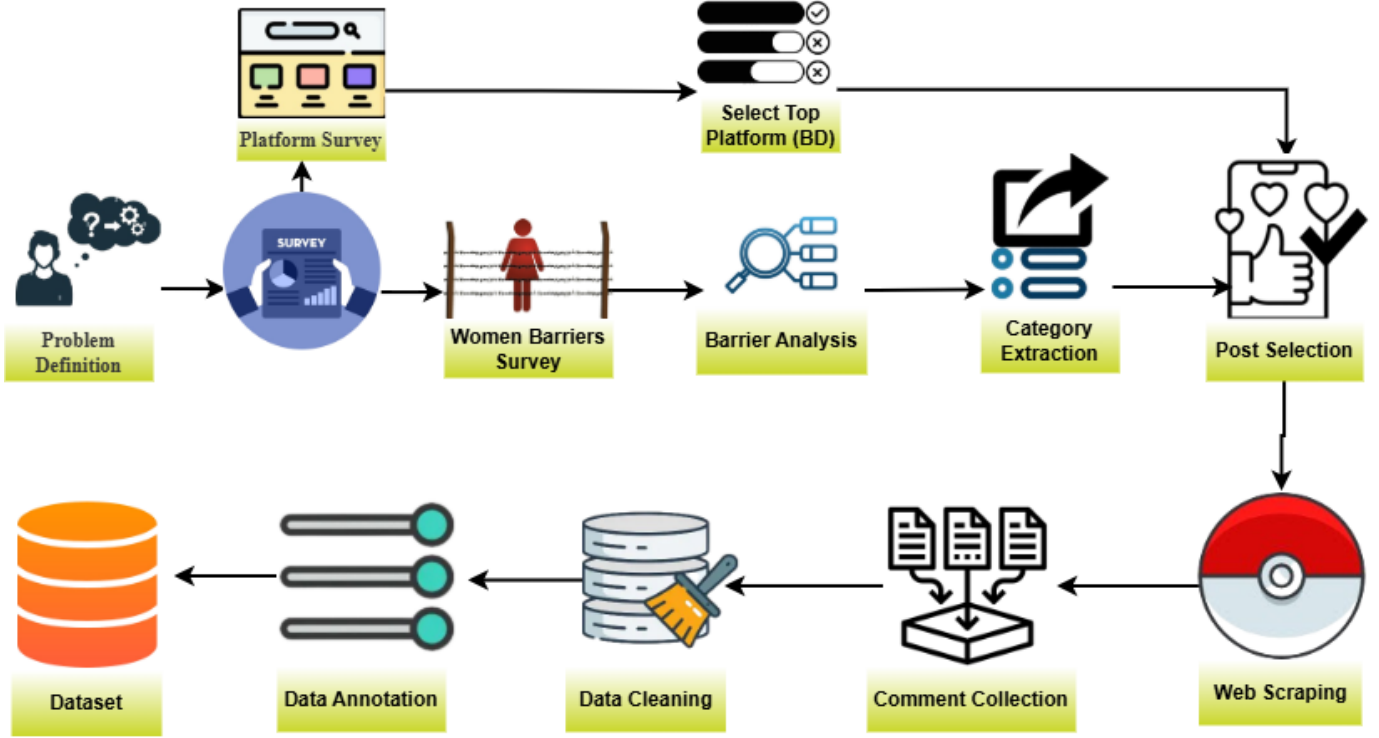


Figure 1. PoWoBD Dataset Construction Pipeline

Economic Barrier, turned out to be rare in the collected comments: family pressure and financial hardship are things women experience directly within their families, not the kind of thing strangers write about in comments on a woman politician’s posts. The dataset is therefore built around the remaining seven labels.

C. Data Collection

For collecting comments, we are targeting Bangladeshi women who are directly or indirectly involved in Bangladesh politics. For that reason, we need to specify the social media platforms from which we will collect comments for the PoWoBD dataset. We are conducting another survey with the purpose of selecting the most used social media platform in Bangladesh and the platform most frequently associated with cyberbullying, online harassment, online threats, and negative comments toward women politicians. The response to the survey for platform selection is below: From figure-2 we can clearly understand that the most used social media is Facebook in Bangladesh. After Selecting Facebook as source of data collection, dividing the full process of data collection into several steps. The steps explain below:

Post Selection: We are selecting various types of Facebook posts from news media pages, political party

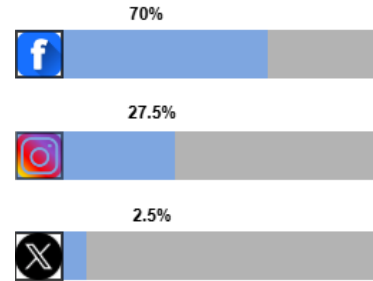


Figure 2. Result of Platform Survey in percentage

pages, individual women politicians pages, feminist pages, public groups whose posts are directly or indirectly aligned with women’s politics or women’s leadership in Bangladesh politics. We collected 100+ posts from different pages and groups whose data are publicly available.

Web scrapping: After post selection, we collected 300-350 comments per post from individuals using Instant Data Scraper of chrome extension. The scrapping process extracts comments with extra information like timestamps, and publicly available metadata of users. All collected data stored in a structured format(CSV)

for further preprocessing. During this extraction process, we ensure that our collected comments must be publicly available.

Data cleaning: Once the web scraping was done, all comments underwent a thorough cleaning process. Removing all the duplicated comments, spam content, irrelevant comments and non-textual content(emojis, URLs, Hashtags, special characters) which are non-relevant with PoWoBD dataset. Finally, the cleaned data was standardized and properly formatted for the next phase of annotation.

D. Annotation Protocol

Our Annotation team comprised four members of university students(two male, two female) who are fluent in standard Bengali. Whereas the team members were selected strategically to represent diversity geographically, culturally and religiously. To ensure this diversity, our team members differ with geographical and cultural perspectives across the four divisions of Bangladesh’s: Chittagong, Khulna, Mymensingh, Sylhet. The team also reflected religious diversity, with three members being Muslim and one member being Hindu. This composition helped to ensure varied perspectives and minimize the bias during the annotation process.

Control Annotation Quality: At the Starting phase of Annotation,a stratified subset of 500 comments was selected to provide representation across the preliminary categories. After the subset annotation was completed, the results were compared and discussed among annotators to identify discrepancies, clarify ambiguities and align on the annotation guidelines. This step was taken to improve consistency and accuracy before moving to annotate the full dataset.

After the completion of annotations of the full dataset, we calculated the inter annotator agreement using Fleiss’ kappa(Fleiss, 1971) and obtained a score of 0.71 which indicates Substantial agreement among the annotators. This score represents the PoWoBD dataset is highly consistent and reliable for more analysis.

E. Dataset Statistics

After completing the previous steps of data constructions, we have a total 23,532[Table 1] instances of data in PoWoBD Dataset. The dataset contains 31.42% of *Neutral* instances which is the largest instance of class label and 3.31% *Gender Discrimination* which is the smallest instance of class label among seven categories. From the class label distribution [figure 3], 31.42% of data instances labeled as *Neutral*, 28.67% *Personal Attack*, 13.82% *Character Shaming*, 12.75% *Supportive*, 5.72% *Personal Safety Threat*, 4.31% *Psychological Impact*, 3.31% *Gender Discrimination*. The *Neutral* and *Personal Attack* labels together cover 60.09%, while *Character*

Labels	Instances	#Sent/Instance	#Words/Instance
Neutral	7394 (31.42%)	1.18	8.63
Personal Attack	6747 (28.67%)	1.15	8.16
Character Shaming	3252 (13.82%)	1.11	7.43
Supportive	3001 (12.75%)	1.20	8.18
Personal Safety Threat	1345 (5.72%)	1.10	8.02
Psychological Impact	1014 (4.31%)	1.30	10.73
Gender Discrimination	779 (3.31%)	1.25	10.94
Total	23532	1.17	8.40

Table 1: Brief statistics of PoWoBD per class label.

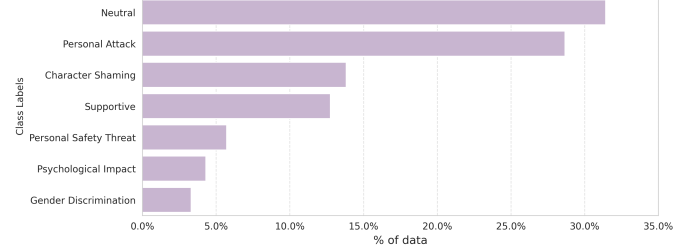


Figure 3. Class label distribution (% of instances).

Shaming and *Supportive* covers 26.57% data instance of full dataset. The remaining 13.34% of data instances are covered by *Personal Safety Threat*, *Psychological Impact*, and *Gender Discrimination*. Therefore, these three classes contain less data instances compared to other class labels. Furthermore, from [table 1] the dataset contains on average 1.17 sentences per instance and 8.40 words per sentence for every class label. However, the *Psychological Impact*, and *Gender Discrimination* classes have more than 10 on average words per sentence which indicates that those class labels contain relatively longer and descriptive comments compared to other classes.

IV. MODEL EVALUATION

A. Experimental Setup

All experiments we are used 90:10 train-test split, maximum tokenized sequence length used 128 tokens. Fine-tuning was carried out with the Hugging Face *Trainer* API, a learning rate of 2e-5, and batch sizes of 16-24 depending on the model, for 3-5 epochs. Four transformer backbones were tested on our dataset which will be described below: *bert-base-multilingual-cased*(mBERT), *csebuetnlp/banglabert*, *csebuetnlp/banglishbert* and *xtlm-roberta-base*.

B. Reducing Label Granularity

The original PoWoBD taxonomy carries seven fine-grained labels, whereas *Character Shaming*, *Psychological Impact*, *Personal Safety Threat*, and *Personal Attack*, for instance, frequently co-occur in the same comment: an insult aimed at a woman’s character often also carries a personal attack, and the two are not always distinguishable from surface wording alone. A preliminary attempt to train directly on the nine-class scheme produced weak and unstable accuracy, which is consistent with this overlap —

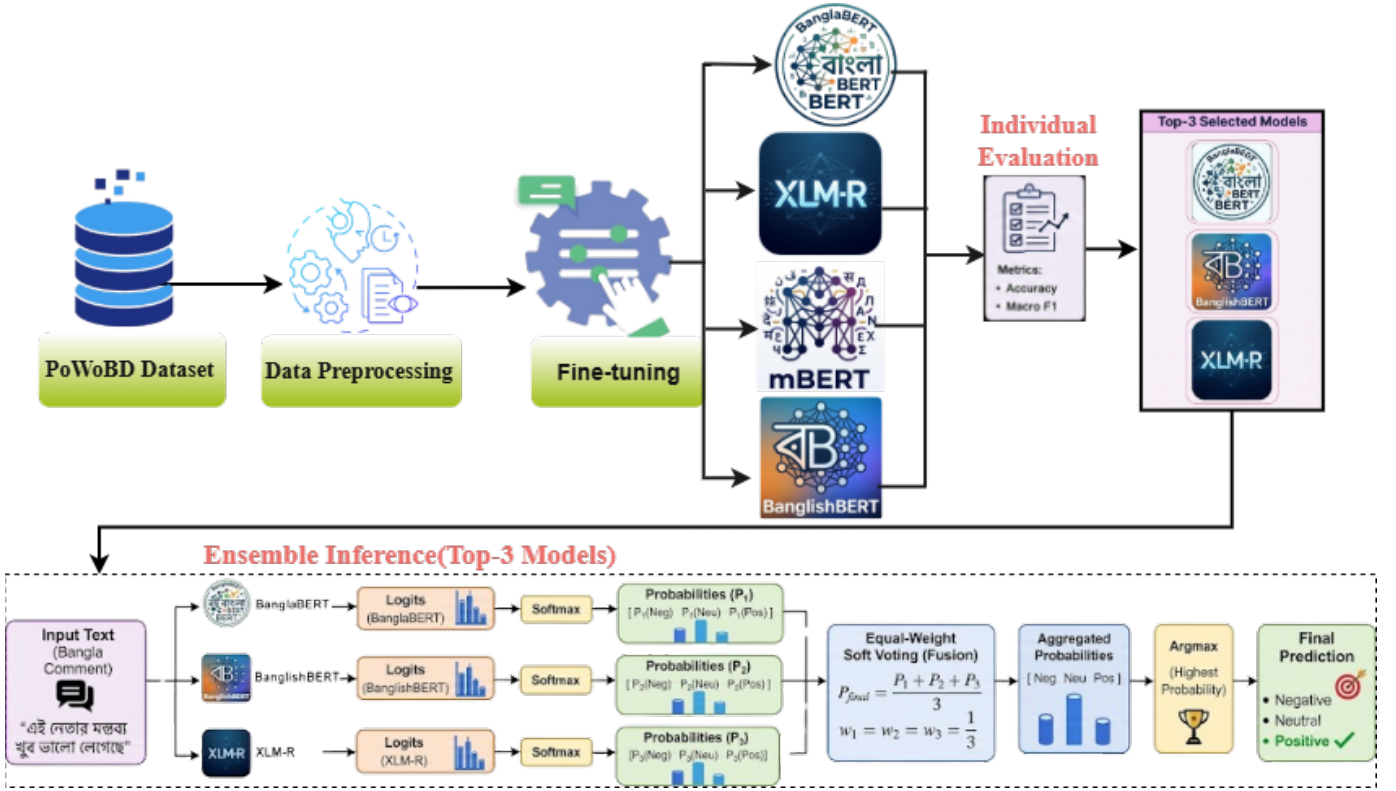


Figure 4. Model Evaluation With PoWoBD Dataset

the model had too many semantically adjacent categories to separate with a limited number of examples per class.

To address this, the label set was consolidated: *Psychological Impact*, *Personal Safety Threat*, and *Character Shaming* were folded into *Personal Attack*, leaving a four-class scheme (*Gender Discrimination*, *Neutral*, *Personal Attack*, *Supportive*). Fine-tuning mBERT on this reduced scheme raised accuracy to 68.03% (macro F1 0.59; Table II), with a repeat run reaching 68.49% (macro F1 0.62). A class-weighted loss, intended to help the minority *Gender Discrimination* class, instead pulled overall accuracy down to 64.82% — recall for that class improved slightly (0.40 to 0.43) but at a real cost to precision (0.43 to 0.28), so the trade-off did not pay off in aggregate.

Table II
MBERT PERFORMANCE ON THE FOUR-CLASS LABEL SCHEME.

Class	Precision	Recall	F1	Support
Gender Discrimination	0.43	0.40	0.42	82
Neutral	0.60	0.58	0.59	768
Personal Attack	0.74	0.77	0.76	1209
Supportive	0.71	0.69	0.70	312
Accuracy	0.6849			
Macro avg	0.62	0.61	0.62	2371

Even at four classes, *Gender Discrimination* remained the weakest category by a wide margin (F1 0.42 against 0.59-0.76 for the others), which reflects both its small support (82 test instances) and its conceptual overlap with

Personal Attack — a comment can express a patriarchal attitude and a direct insult at the same time, and the boundary between the two is often a matter of degree rather than kind.

C. Collapsing to Three-Class Polarity Scheme

Table III
REPRESENTATIVE DISAGREEMENTS BETWEEN THE ORIGINAL LABEL AND THE FOUR-CLASS MODEL’S PREDICTION.

Comment (Bangla)	True Label	Model Label
তিলোত্তমা শিকদার বাংলাদেশ গর্ভলীগ (Tilottama Shikder, Bangladesh 'Gorto League' – a vulgar pun replacing Awami League with “ ” (a crude slang for a body part), mocking her as part of that “league.”)	Personal Attack	Gender Dis- crimination
মেয়েরা বরাবরই বাতিলের খাতায়। শুধু শুধু চেষ্টা করা আরকি। একা এভাবে আগানো প্রায় অসম্ভব.. (Girls are always written off / dismissed as rejects. It's basically a pointless effort. Moving forward alone like this is almost impossible..)	Gender Dis- crimination	Personal Attack

Manual inspection of the four-class model’s errors showed that most of its mistakes were not random but systematic: comments originally labeled under the fine-grained categories *Personal Safety Threat*, *Character Shaming*, *Personal Attack*, and *Gender Discrimination*

were frequently predicted as one another, particularly as *Gender Discrimination*, regardless of which category the comment actually belonged to (Table III). This indicates that the model itself was unable to reliably separate these categories from the surface text, which is consistent with the class-level results in Table II, where *Gender Discrimination* had the lowest F1 of any class.

Table IV
ACCURACY AND MACRO-F1 ACROSS MODELS ON THE THREE-CLASS SCHEME.

Model	Accuracy	Macro F1
mBERT	77.39%	0.7395
XLM-RoBERTa	78.74%	0.7530
BanglishBERT	80.43%	0.7746
BanglaBERT (weighted)	76.80%	0.74
BanglaBERT	82.12%	0.7896

Rather than continuing to force a boundary the model could not consistently draw, the four classes were mapped onto three sentiment-level classes: *Supportive* became *positive*, *Neutral* stayed *neutral*, and *Personal Attack* together with *Gender Discrimination* were merged into a single *negative* class. The resulting dataset was used to fine-tune all four backbones.

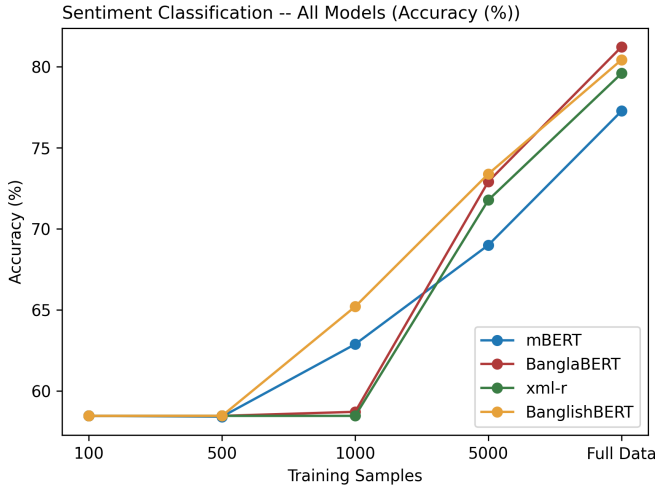


Figure 5. All Models Accuracy

BanglaBERT gave the best result among the individually fine-tuned models, reaching 82.12% accuracy with a macro F1 of 0.7896. *Negative* comments, which dominate the dataset, were classified most reliably across every model (F1 0.84-0.88), while *neutral* remained the weakest class throughout (F1 0.64-0.70) — consistent with the earlier four-class run and with what the BLP-2023 shared task reported for Bangla sentiment classification more broadly [3], where a genuinely neutral comment and one carrying faint, indirect sentiment are not always easy to tell apart even for annotators. BanglishBERT, trained on the same Electra-based architecture as BanglaBERT but pretrained on code-mixed Bangla-English text, slotted in

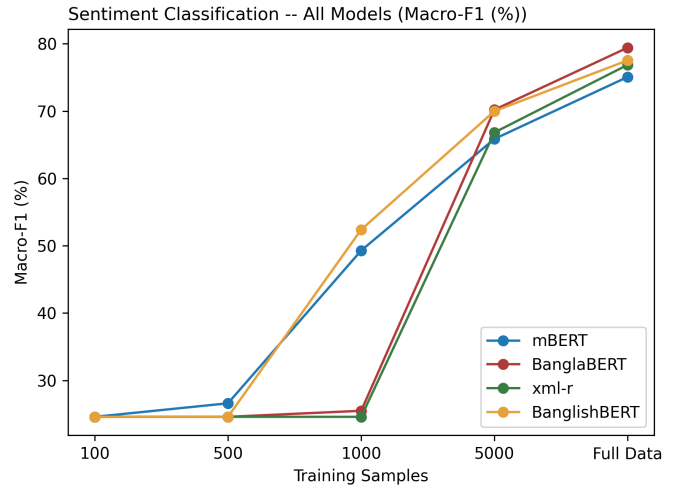


Figure 6. All Models Macro f1

between mBERT/XLM-R and BanglaBERT, suggesting that some of BanglaBERT’s advantage comes from architecture rather than the Bangla-only pretraining corpus alone. Class-weighted training, tried on BanglaBERT as a response to the residual imbalance between negative and positive comments, again underperformed the unweighted model (76.80% vs. 82.12% accuracy), reinforcing that reweighting the loss was not an effective strategy for this dataset at either the four-class or three-class stage.

D. Ensemble of Fine-tuned Models

The four fine-tuned models were initially considered for ensemble construction. Based on macro-F1, mBERT was excluded, and the remaining three models—BanglaBERT, BanglishBERT, and XLM-R—were combined. Two weighting schemes were compared over the remaining three models (BanglaBERT, BanglishBERT, XLM-R): equal weighting, and weighting each model’s contribution in proportion to its individual macro-F1. Performance-weighting performed marginally better (accuracy 83.76%, macro F1 0.8160) than Equal weighting (accuracy 83.24%, macro F1 0.8140), which suggests that once the three retained models are already close in quality, tilting the vote toward the single best one adds little.

Table V
BEST ENSEMBLE (PERFORMANCE-WEIGHT)

Class	Precision	Recall	F1	Support
Negative	0.8631	0.9093	0.8856	1400
Neutral	0.7834	0.6869	0.7320	674
Positive	0.8197	0.8418	0.8306	297
Accuracy			0.8376	
Macro avg	0.8220	0.8127	0.8160	2371
Weighted avg	0.8350	0.8376	0.8350	2371

The ensemble improved on BanglaBERT alone by nearly a full percentage point of accuracy (83.76% vs. 82.12%) and by a comparable margin in macro F1 (0.8160

vs. 0.7896), with the largest single gain in negative-class recall (0.9093, up from BanglaBERT’s 0.89). The confusion matrix for the ensemble shows that most of the remaining error is concentrated in the neutral class being predicted as negative rather than in confusion between the two sentiment poles, which mirrors the pattern already seen in every individual model and points to neutral-vs-negative boundary cases — comments that are mildly critical without being overtly hostile — as the dataset’s hardest cases rather than any weakness specific to one architecture.

E. Discussion

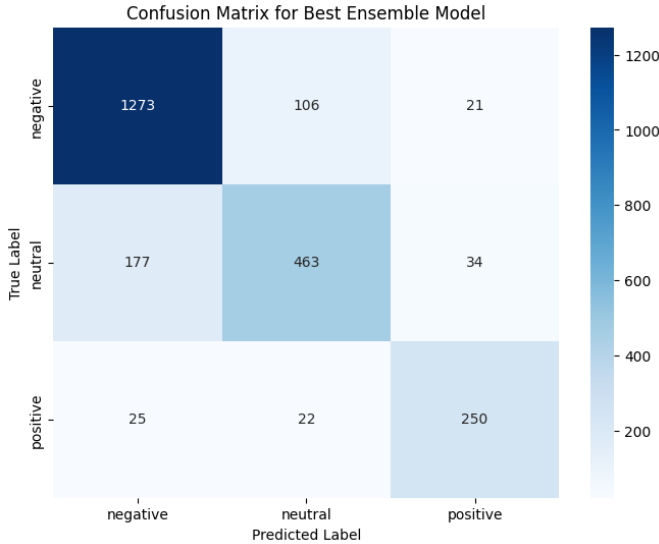


Figure 7. Confusion Matrix

Looking across the three modeling stages together, a clear pattern emerges: performance did not improve because the comments became easier to classify, but because the label space stopped asking the model to draw distinctions that Bengali political discourse itself does not draw cleanly. *Character Shaming*, *Psychological Impact*, and *Personal Attack* were designed as separate categories during taxonomy design (Section III-B), and they are separate in principle, but a single hostile comment toward a woman politician tends to carry several of these at once. The annotators’ own agreement score (Fleiss’ $\kappa = 0.71$) already hinted that even trained humans found some of these boundaries harder to hold than others, and the four-class model’s confusion pattern (Table III) made that difficulty concrete rather than theoretical: it was not failing randomly, it was failing exactly where a human reader might also hesitate.

The move to three sentiment-level classes relocated rather than removed this difficulty. *Neutral* and *negative* became the hardest boundary, since a mildly critical comment and a neutral one often look identical on the surface until read past the first clause — consistent with

what BLP-2023 also found for Bangla sentiment more broadly [3], suggesting the difficulty belongs to Bangla political commentary as a genre rather than to this dataset specifically.

The ensemble result fits this picture rather than standing apart from it. No single backbone was uniformly best across all three classes — BanglaBERT handled *negative* and *neutral* more reliably, while BanglishBERT and XLM-RoBERTa were competitive on *positive* — suggesting the models picked up on partly different signals from their different pretraining corpora. Averaging their predictions let the ensemble draw on whichever model was more confident for a given class, concentrating the gain in *negative*-class recall. That the gain was modest rather than dramatic further confirms the remaining error is a genuine ambiguity in the comments, not a modeling gap.

Two practical conclusions follow from this trade-off between the two label schemes. Because the nine-class taxonomy preserves the distinction between categories such as character attacks and safety threats — a distinction that carries real consequence, since the appropriate response to a threat against someone’s physical safety is not the same as the appropriate response to reputational harm — it remains the more useful representation for the qualitative and policy-facing goals in Section V, even though it is harder for a model to classify reliably on its own. The three-class ensemble, by contrast, sacrifices that granularity but gains something a deployed system needs more: predictable failure. Its errors are not scattered unpredictably across categories the way the nine-class model’s were; they concentrate specifically at the neutral-negative boundary, where the ambiguity lies in the comment itself rather than in the model’s reasoning. This predictability is what makes it the more defensible choice for systems that must act on comments as they arrive — borderline cases near that boundary can be routed to human review as a matter of course, rather than the system’s confidence being trusted uniformly across all cases, including ones where the model has no particular reason to be right.

V. APPLICATION

The proposed dataset has multiple applications in computational, social, and policy-making domains. Firstly, this dataset can help in automatic detection of politically gender-based harassment against Bangladeshi women on social media. Here, we annotated the dataset into seven fine-grained classes so that machine learning models can identify the personal attack, character shaming, personal safety threat, gender discrimination, psychological impact targeting. Secondly, it can serve as an important resource for content moderation systems. Currently, the number of high-quality and annotated resources for detecting gender-based political harassment in the Bangla language is very limited. In this context, the proposed dataset can fill an important gap. Social media platforms and digital governance take initiatives and they can use models trained

on this dataset to detect and manage offensive, hateful, or discriminatory content targeting women in politics in real-time more effectively.

Thirdly, the dataset can be used in the analysis of bias and political discourse analysis. Researchers will be able to analyze the patterns of hostile narratives, structural barriers, and supportive statements surrounding women’s political participation in Bangladesh. This will make a significant contribution to research on gender-based political communication in low-resource language contexts. Finally, this dataset provides evidence-based insights for policymaking and advocacy activities. Policymakers, civil society organizations, and groups related to women’s rights can use insights from this dataset to take effective steps to reduce online harassment and address the barriers that discourage women’s political participation.

VI. CONCLUSION AND FUTURE WORK

This work introduced PoWoBD, a survey-grounded Bangla dataset targeting the barriers women face in Bangladeshi political leadership, and evaluated it under three progressively broader label schemes. The best-performing configuration, a soft-voting ensemble of BanglaBERT, BanglishBERT, and XLM-RoBERTa fine-tuned on the three-class (positive/negative/neutral) re-annotation, reached 83.76% accuracy and a macro F1 of 0.8160, improving accuracy by 1.64 percentage points over BanglaBERT.

In the future work, the dataset can be expanded and more comments can be collected from different social media platforms. This would increase the generalizability of the dataset. Moreover The original nine class label can further be refined by the original nine-category taxonomy can further be refined by collecting additional comments for underrepresented categories such as Gender Discrimination and by investigating whether Family Restriction and Economic Barrier can be reliably represented in social-media comments. because these categories have fewer examples than the other categories. This would make analysis of the categories of the more accurate. Future research can also include explainable AI techniques so that it can be understandable that why a model assigns a specific label to a comment. This can improve the transparency of the system.

REFERENCES

- [1] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “Bert: Pre-training of deep bidirectional transformers for language understanding,” in *Proceedings of NAACL-HLT*, 2019, pp. 4171–4186. [Online]. Available: <https://aclanthology.org/N19-1423/>
- [2] S. Saha, K. M. N. Asib, and M. M. Hoque, “Retriv at blp-2025 task 1: A transformer ensemble and multi-task learning approach for bangla hate speech identification,” *arXiv preprint arXiv:2511.07304*, 2025. [Online]. Available: <https://arxiv.org/abs/2511.07304>
- [3] M. A. Hasan, F. Alam, A. Anjum, S. Das, and A. Anjum, “Blp-2023 task 2: Sentiment analysis,” in *Proceedings of the First Workshop on Bangla Language Processing (BLP-2023)*, Singapore, 2023, pp. 354–364. [Online]. Available: <https://aclanthology.org/2023.banglalp-1.48/>
- [4] K. Fatema, U. Das, M. A. Mia, M. S. Mowla, M. Yahan, M. F. Ullah, A. Sarker, and H. Murad, “Emptymind at blp-2023 task 2: Sentiment analysis of bangla social media posts using transformer-based models,” in *Proceedings of the First Workshop on Bangla Language Processing (BLP-2023)*, Singapore, 2023, pp. 300–304. [Online]. Available: <https://aclanthology.org/2023.banglalp-1.39/>
- [5] K. I. Islam, S. Kar, M. S. Islam, and M. R. Amin, “Sentnob: A dataset for analysing sentiment on noisy bangla texts,” in *Findings of the Association for Computational Linguistics: EMNLP 2021*, Punta Cana, Dominican Republic, 2021, pp. 3265–3271. [Online]. Available: <https://aclanthology.org/2021.findings-emnlp.278/>
- [6] M. R. I. Aupi, N. Tafannum, M. S. Rahman, K. M. Hassan, and N. Rahman, “Wonbias: A dataset for classifying bias prejudice against women in bengali text,” in *Proceedings of the 6th Workshop on Gender Bias in Natural Language Processing (GeBNLP)*, Vienna, Austria, 2025, pp. 105–110. [Online]. Available: <https://aclanthology.org/2025.gebnlp-1.10/>
- [7] M. Kabir, O. B. Mahfuz, S. R. Raiyan, H. Mahmud, and M. K. Hasan, “Banglabook: A large-scale bangla dataset for sentiment analysis from book reviews,” in *Findings of the Association for Computational Linguistics: ACL 2023*, Toronto, Canada, 2023, pp. 1237–1247. [Online]. Available: <https://aclanthology.org/2023.findings-acl.80/>
- [8] M. E. Islam, L. Chowdhury, F. A. Khan, S. Hossain, M. S. Hossain, M. M. O. Rashid, N. Mohammed, and M. R. Amin, “Sentigold: A large bangla gold standard multi-domain sentiment analysis dataset and its evaluation,” in *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2023, pp. 4207–4218. [Online]. Available: <https://arxiv.org/abs/2306.06147>
- [9] Y. Li, Z. Lu, F. Tang, S. Lai, M. Hu, Y. Zhang, H. Xue, Z. Wu, I. Razzak, Q. Li, and J. Su, “Rhythm of opinion: A hawkes-graph framework for dynamic propagation analysis,” *arXiv preprint arXiv:2504.15072*, 2025. [Online]. Available: <https://arxiv.org/abs/2504.15072>
- [10] J. Hartanto, T. Liundi, R. Sutoyo, and E. W. Andangsari, “Id-smsa: Indonesian stock market dataset for sentiment analysis,” *Data in Brief*, vol. 60, 2025. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2352340925003038>
- [11] Z. Wei and S. Zhang, “A structured sentiment analysis dataset based on public comments from various domains,” *Data in Brief*, 2024. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2352340924002038>
- [12] A. Islam, S. K. Roy, and M. E. Hamid, “Bengalisent140: A large-scale bengali binary sentiment dataset for hate and non-hate speech classification,” *arXiv preprint arXiv:2601.20129*, 2026. [Online]. Available: <https://arxiv.org/abs/2601.20129>
- [13] T. Azhar, T. Mahmud, M. A. Hasan, M. N. Uddin, and S.-B. Park, “Hatebertbn: A hybrid transformer based model for bangla hate speech detection across various social contexts,” 2025. [Online]. Available: https://www.researchgate.net/publication/399583775_HateBertBN_a_hybrid_transformer_based_model_for_Bangla_hate_speech_detection_across_various_social_contexts
- [14] M. S. Başarslan and F. Kayaalp, “Sentiment analysis with ensemble and machine learning methods in multi-domain datasets,” *Turkish Journal of Engineering*, vol. 7, no. 2, 2023. [Online]. Available: <https://dergipark.org.tr/en/download/article-file/2277406>
- [15] T. Davidson, D. Warmley, M. Macy, and I. Weber, “Automated hate speech detection and the problem of offensive language,” in *Proceedings of the Eleventh International AAAI Conference on Web and Social Media (ICWSM)*, 2017, pp. 512–515. [Online]. Available: <https://arxiv.org/abs/1703.04009>
- [16] N. R. Bhowmik, M. Arifuzzaman, and M. R. H. Mondal, “Sentiment analysis on bangla text using extended lexicon dictionary and deep learning algorithms,” *Array*, vol. 13, 2022.

- [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S259000562100059X>
- [17] F. Haider, F. T. Shifat, M. F. Ishmam, M. S. U. R. Sourove, D. D. Barua, M. Fahim, and M. F. A. Bhuiyan, "Banth: A multi-label hate speech detection dataset for transliterated bangla," in *Findings of the Association for Computational Linguistics: NAACL 2025*. Albuquerque, New Mexico: Association for Computational Linguistics, 2025, pp. 7232–7251. [Online]. Available: <https://aclanthology.org/2025.findings-naacl.403/>
 - [18] M. Das, S. Banerjee, P. Saha, and A. Mukherjee, "Hate speech and offensive language detection in bengali," in *Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing*, 2022, pp. 286–296. [Online]. Available: <https://aclanthology.org/2022.aacl-main.23/>
 - [19] N. Romim, M. Ahmed, M. S. Islam, A. S. Sharma, H. Talukder, and M. R. Amin, "Bd-shs: A benchmark dataset for learning to detect online bangla hate speech in different social contexts," in *Proceedings of the 13th Conference on Language Resources and Evaluation (LREC 2022)*, Marseille, France, 2022, pp. 5153–5162. [Online]. Available: <https://aclanthology.org/2022.lrec-1.552/>
 - [20] M. A. Salam, S. Islam, M. M. Uddin, T. R. Shraboni, A. Das, M. M. Islam, and M. R. Karim, "Identifying influential determinants of women's empowerment in bangladesh using machine learning algorithms," *PLOS ONE*, 2025. [Online]. Available: <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0338037>
 - [21] R. Ahmed, "Breaking barriers: Women empowerment transforming societies in modern times building a better future," *Current Trends in Mass Communication*, vol. 3, no. 1, pp. 1–12, 2024. [Online]. Available: https://www.preprints.org/frontend/manuscript/a4c83a2d93704df0f9b480ff3c09c14e/download_pub
 - [22] F. Ara and J. Northcote, "Women's participation in bangladesh politics, the gender wall and quotas," *South Asia Research*, 2020. [Online]. Available: <https://doi.org/10.1177/0262728020915562>